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# Data-Centric Multimodal Pavement Distress Detection Using Intensity–Range Imagery

**Presenter:**
**Wenhan Tao**
PhD Student, Department of Computer Science
Texas State University

**Advisors:**
**Dr. Jelena Tešić, Dr. Feng Wang, Dr. Yongsheng Bai**

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**GitHub Repository:**
https://github.com/DataLab12/independent-study-icmr-2026

# Presentation Roadmap

1. Motivation / Problem
2. TxDOT Intensity–Range Dataset
3. Proposed Method
4. Experimental Setup
5. Results
6. Conclusion / What I Tried and Lessons Learned

# Motivation / Problem

# Motivation: Why Intensity–Range Multimodal Detection?

- Pavement distress detection is important for large-scale roadway condition monitoring.
- 2D intensity images capture surface appearance, texture, and visual contrast.
- 3D range images capture surface geometry, depth variation, and structural cues.
- Different distress classes may rely on different modalities.
- The key problem is how to combine intensity and range without losing class-critical information.

# Example: Paired Intensity and Range Images

2D Intensity Image

3D Range Image

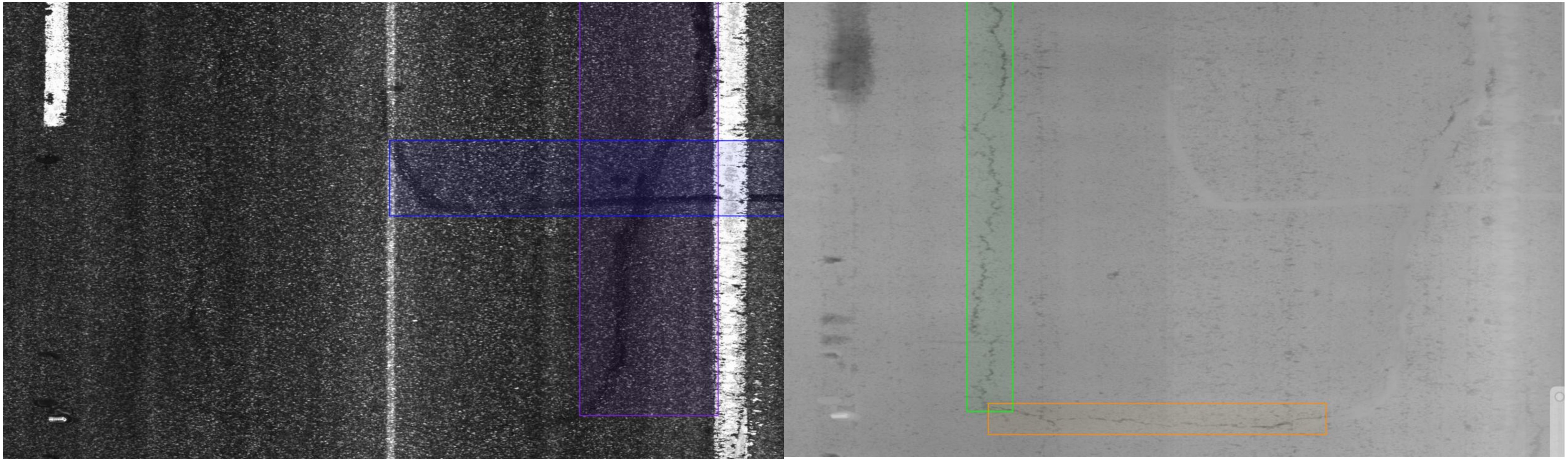

Figure 1: Paired intensity and range images provide complementary 2D appearance and 3D surface cues.

# Research Objective

## Main Goal

- Improve pavement distress detection by better leveraging complementary information from 2D intensity and 3D range imagery.

## Objectives

- 1. Analyze class-wise modality preference between intensity and range.
- 2. Design a split-backbone YOLOv5s model to preserve modality-specific features.
- 3. Develop class- and modality-aware augmentation for paired 2D–3D inputs.
- 4. Evaluate the proposed method against early-fusion and single-modality baselines.

# TxDOT Intensity–Range Dataset

# Dataset Overview

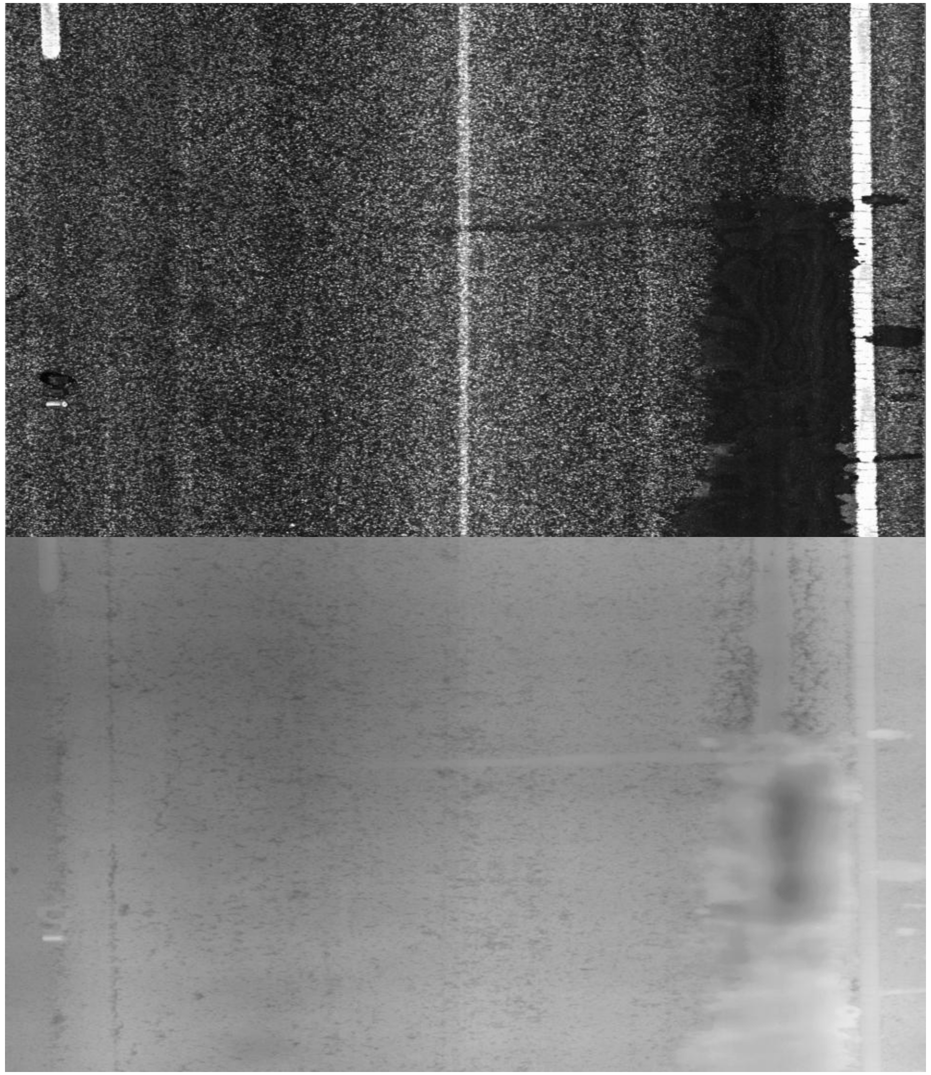

Figure 2: Paired 2D intensity and 3D range images from the TxDOT ACP dataset.

## Dataset Source

- TxDOT 2D/3D Asphalt Concrete Pavement (ACP) dataset
- Collected by a pavement inspection vehicle
- Each sample contains spatially aligned intensity and range images

## Image Modalities

- 2D intensity images: camera-based pavement surface appearance
- 3D range images: laser-based pavement surface measurements
- Aligned 2D/3D images are tiled into 1536 × 900 patches for detection annotation and model training.

# Class distribution

- The TxDOT ACP subset contains nine pavement distress classes.
- The class distribution is highly imbalanced.
- Frequent classes include sealed longitudinal, sealed transverse, longitudinal, and lane longitudinal.
- Rare classes include pothole, block, and alligator.
- Class imbalance makes multimodal detection more challenging, especially for rare distress types.

**Table 1: Class distribution of the TxDOT ACP dataset.**

| Class | # Images | # Instances |
|---|---|---|
| Transverse | 1,183 | 1,673 |
| Joint | 369 | 488 |
| Sealed transverse | 1,506 | 2,101 |
| Longitudinal | 1,369 | 1,853 |
| Lane longitudinal | 1,383 | 1,487 |
| Sealed longitudinal | 2,411 | 4,025 |
| Block | 293 | 296 |
| Alligator | 678 | 878 |
| Pothole | 115 | 154 |
| **Total** | **6,936** | **12,945** |

# Proposed Method

# Proposed Multimodal Detection Workflow

- Input: paired 2D intensity and 3D range images
- Shared geometric augmentation preserves 2D–3D alignment
- Modality-specific pixel augmentation handles intensity and range differently
- Split-backbone YOLOv5s keeps modality features separated in the backbone
- Neck/head performs late fusion and outputs class, box, and confidence

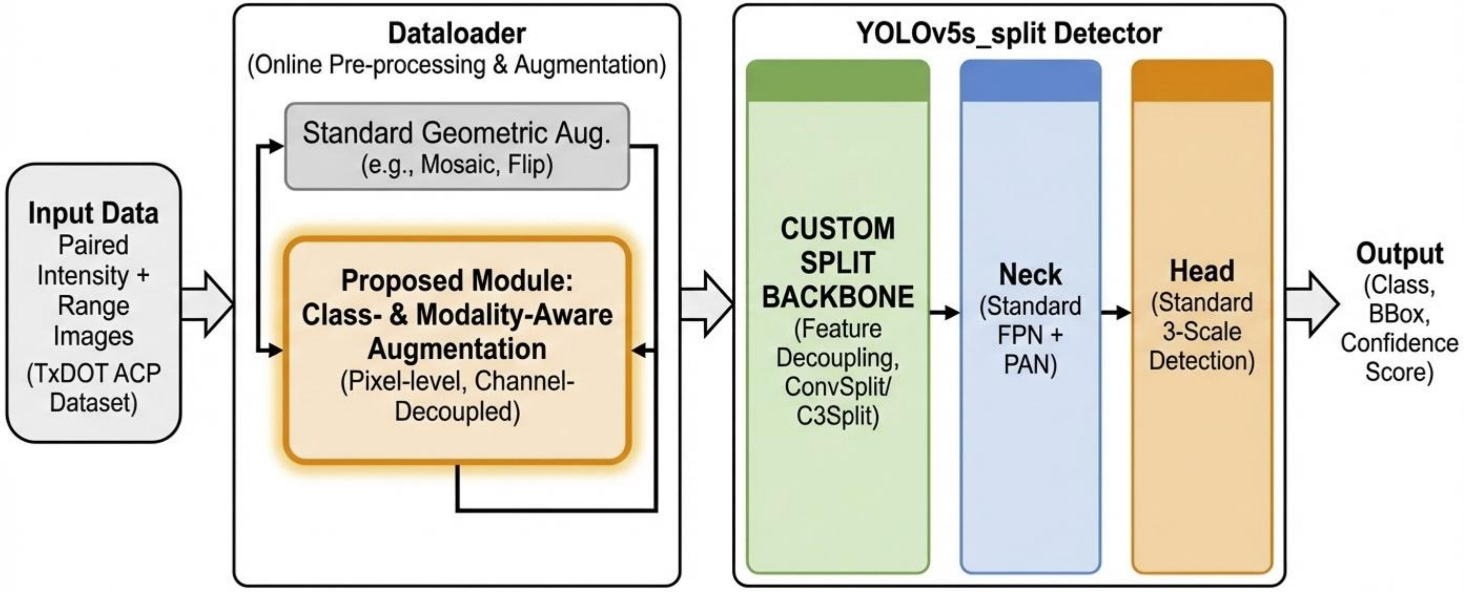


**Figure 4: Proposed intensity–range detection workflow with modality-aware augmentation and split-backbone YOLOv5s.**

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# Step 1 — Dual-Channel Data Pipeline

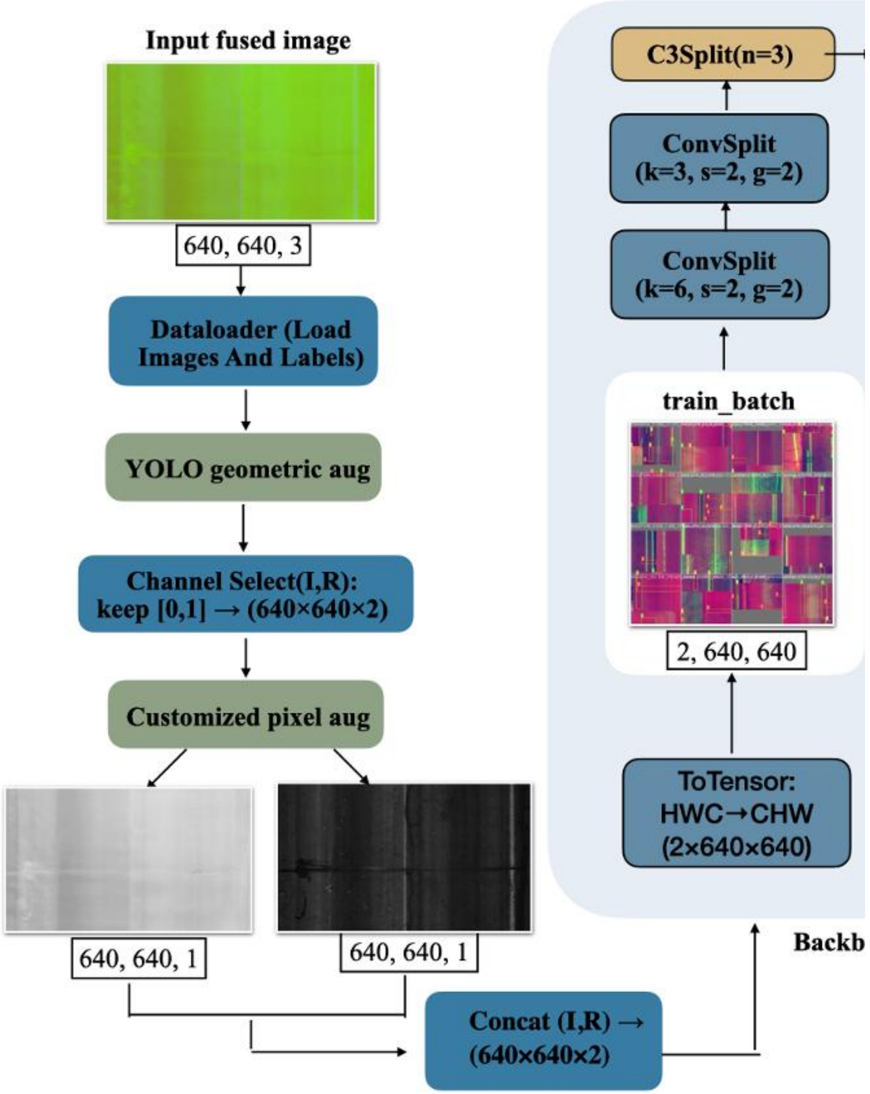



**Figure 4: Dual-channel data pipeline for paired intensity–range inputs.**

- Input image stores intensity, range, and a dummy channel
- Keep only intensity and range channels: [0, 1]
- Shared geometric augmentation preserves 2D–3D alignment
- Pixel-level augmentation is applied separately to each modality
- Final model input: 640 × 640 × 2 tensor

# Step 2 — Split-Backbone YOLOv5s

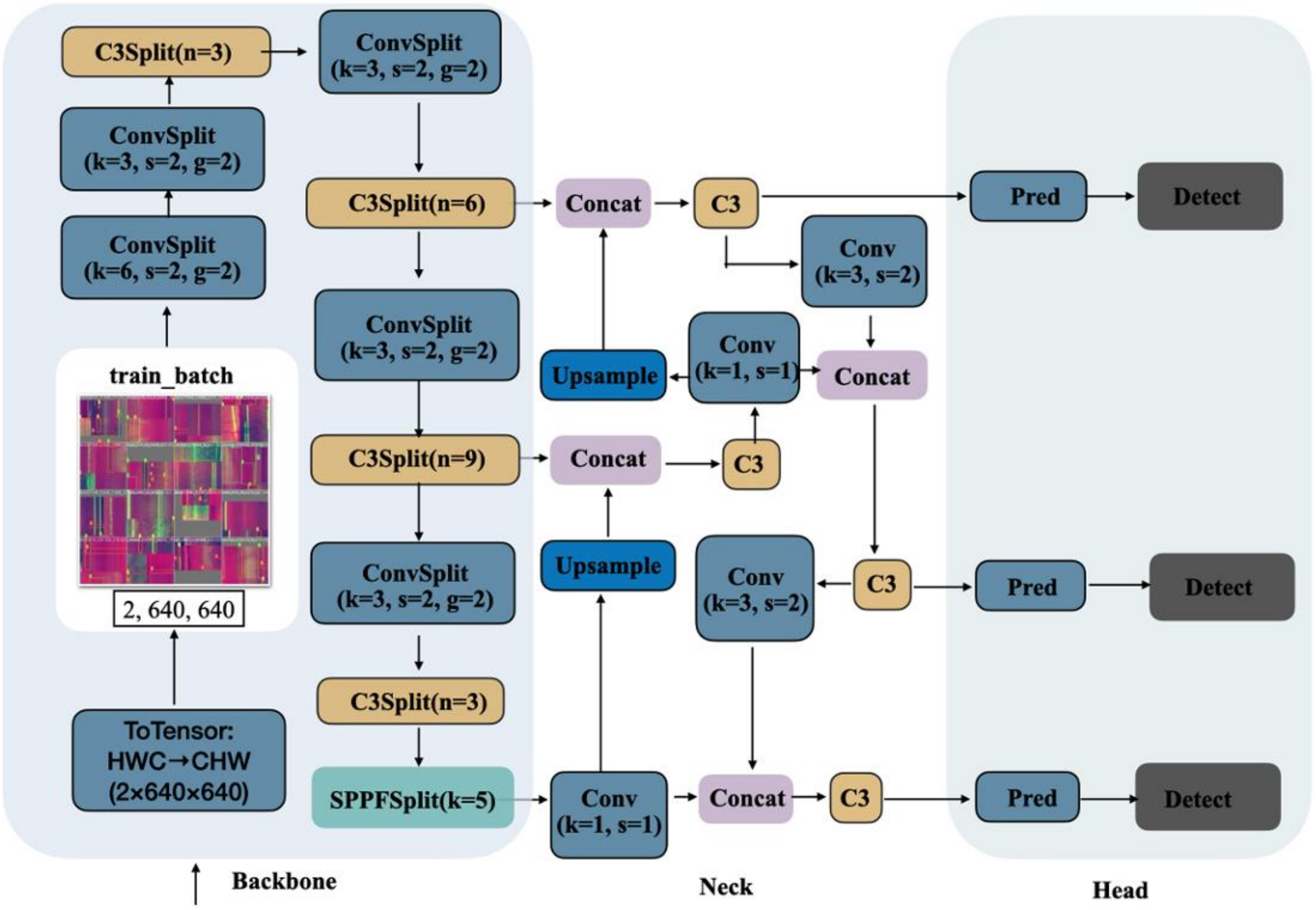



**Figure 5: Split-backbone YOLOv5s architecture for intensity–range pavement distress detection.**

## Key design:

- Only the backbone is modified.
- Backbone Conv, C3, and SPPF blocks are replaced with ConvSplit, C3Split, and SPPFSplit.
- Intensity and range streams remain separated in the backbone.
- Neck and head follow the standard YOLOv5s design.
- Fusion happens after backbone feature extraction.

# Backbone Components: ConvSplit, C3Split, and SPPFSplit

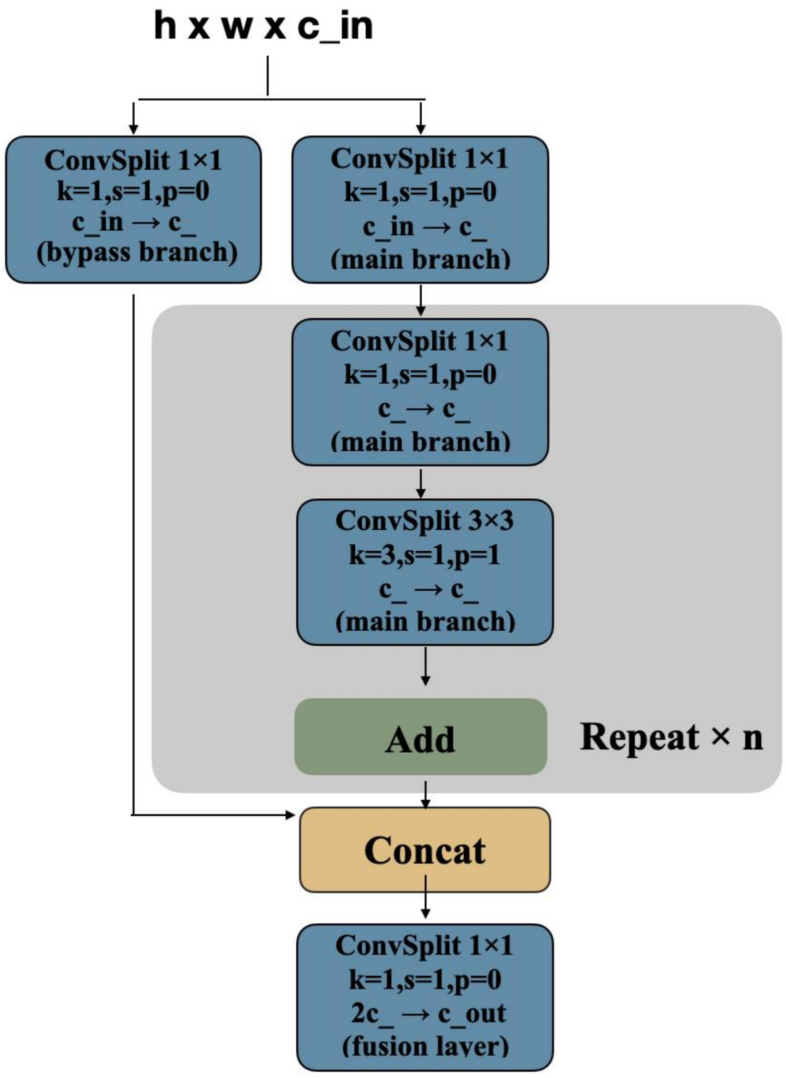



Figure 6: C3Split module replacing standard C3 convolutions with ConvSplit operations.

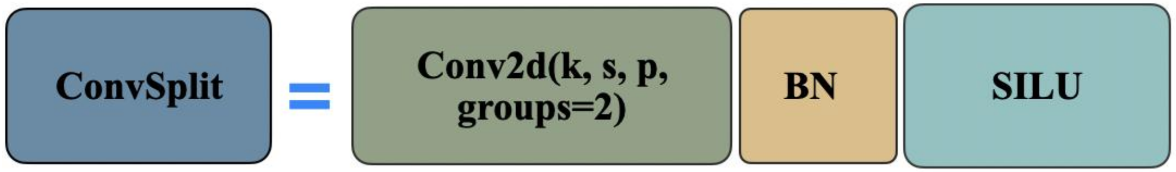



Figure 7: ConvSplit module implemented by grouped convolution with two modality groups.

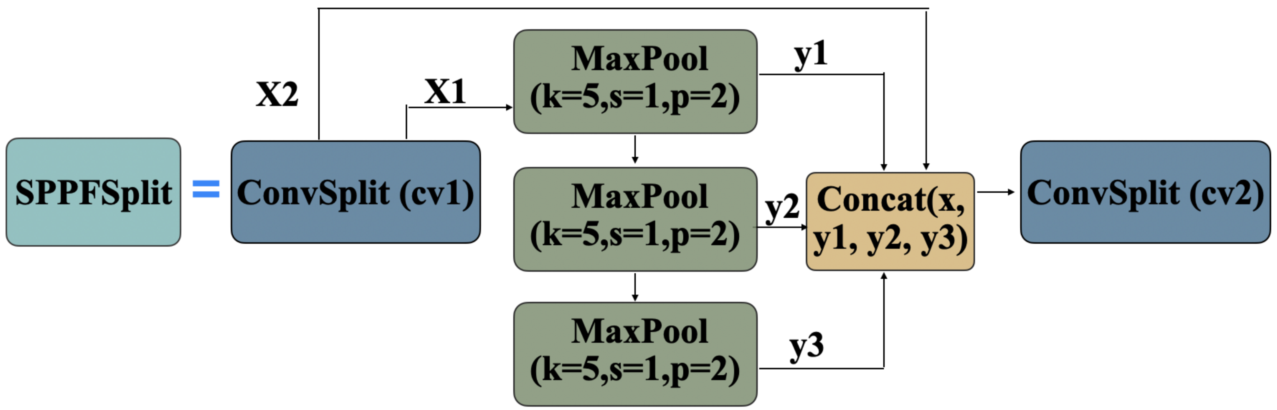



Figure 8: SPPFSplit module preserving multi-scale pooling while using ConvSplit projections.

# Experimental Setup

# Training Setup

**Table 2: Training and evaluation setup for intensity–range YOLOv5s detection.**

| Item | Setting |
| --- | --- |
| Task | ACP pavement distress detection |
| Dataset | TxDOT intensity–range ACP dataset |
| Models | YOLOv5s variants |
| Baselines | I-only, R-only, Early Fusion |
| Proposed | Split-backbone YOLOv5s + designed augmentation |
| Input size | 640 × 640 |
| Epochs | 250 |
| Metrics | AP@0.5 / AP@0.5:0.95 |

# Results

# Test Set Results

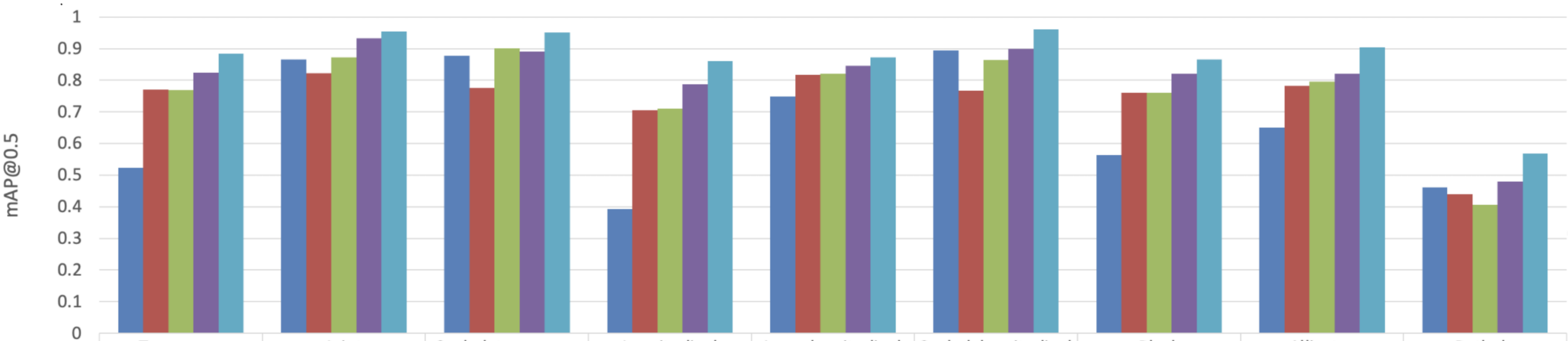
## Key Findings on Test Set

- Proposed method remains effective on the held-out test set.
- Early Fusion 2ch performs worse, so the gain is not from simply reducing channels.
- Split Backbone improves performance by delaying modality fusion.
- Designed augmentation performs best by using class- and modality-aware training.

**Table 3: Overall performance on the held-out TxDOT ACP test set.**

| Method | AP@0.5 | AP@0.5:0.95 | Precision | Recall |
|---|---|---|---|---|
| Early Fusion 3ch | 0.775 | 0.391 | 0.781 | 0.724 |
| Early Fusion 2ch | 0.736 | 0.367 | 0.782 | 0.667 |
| Split Backbone 2ch | 0.799 | 0.402 | 0.781 | 0.754 |
| Split Backbone + Uniform Aug | 0.811 | 0.412 | 0.775 | 0.763 |
| **Split Backbone + Designed Aug** | **0.833** | **0.428** | **0.770** | **0.785** |

# Class-wise Modality Preference




| | Transverse | Joint | Sealed_transverse | Longitudinal | Lane_longitudinal | Sealed_longitudinal | Block | Alligator | Pothole |
|---|---|---|---|---|---|---|---|---|---|
| Intensity | 0.524 | 0.865 | 0.877 | 0.393 | 0.749 | 0.895 | 0.563 | 0.651 | 0.462 |
| Range | 0.77 | 0.822 | 0.775 | 0.705 | 0.817 | 0.767 | 0.76 | 0.782 | 0.439 |
| Fused | 0.769 | 0.872 | 0.901 | 0.71 | 0.821 | 0.864 | 0.761 | 0.795 | 0.406 |
| Split | 0.824 | 0.933 | 0.891 | 0.788 | 0.845 | 0.899 | 0.82 | 0.82 | 0.48 |
| Split+Aug | 0.884 | 0.955 | 0.951 | 0.86 | 0.872 | 0.961 | 0.866 | 0.904 | 0.569 |

**Figure 6: Class-wise AP@0.5 comparison under intensity-only, range-only, early-fusion, split-backbone, and proposed settings.**

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# Qualitative Results

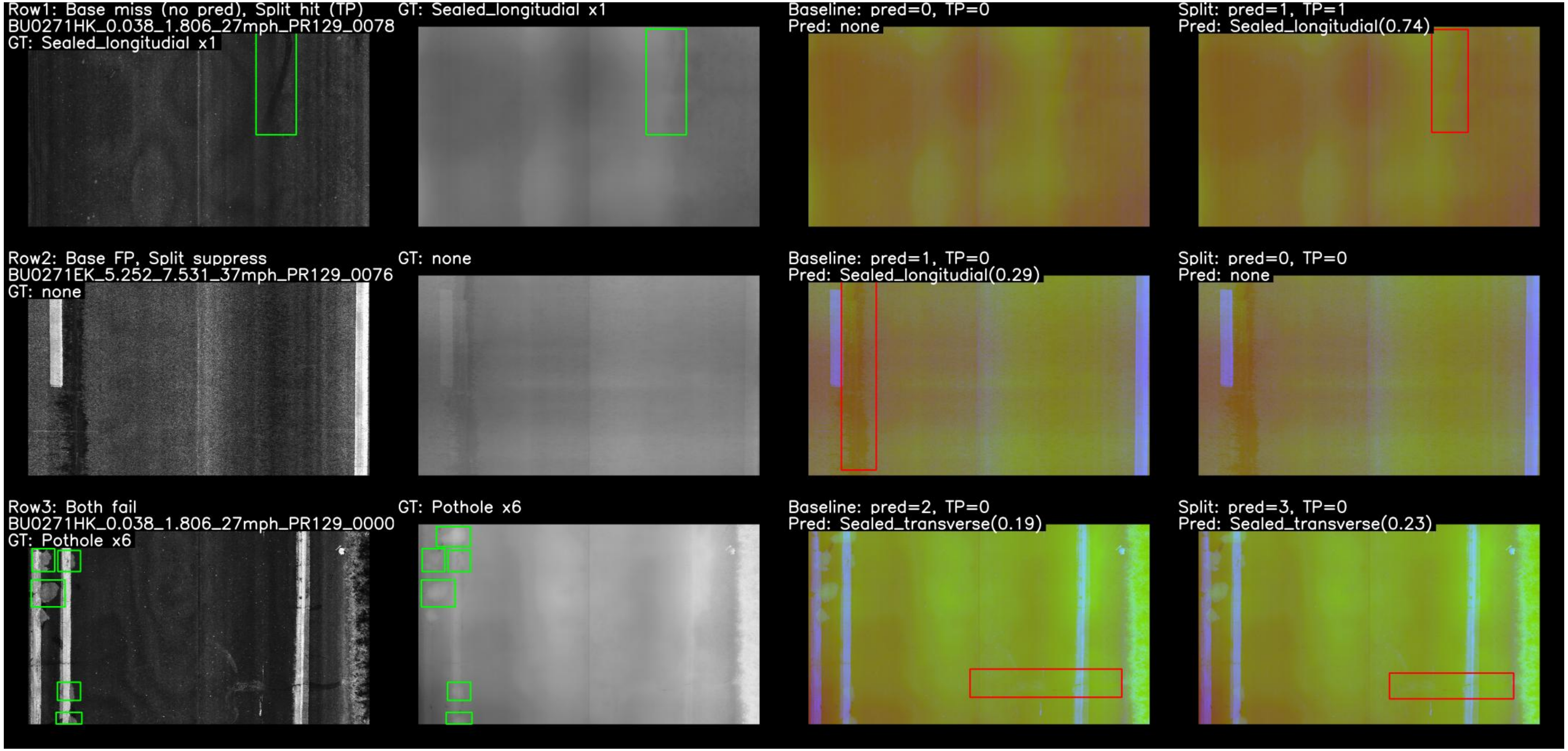


**Figure 7: Qualitative examples comparing intensity image, range image, early-fusion baseline prediction, and proposed model prediction.**

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# Conclusion / What I Tried and Lessons Learned

# Conclusion: Key Findings

- Early fusion is simple, but it can mix modality signals too early.
- Intensity and range provide complementary but class-dependent evidence.
- Split-backbone learning helps preserve modality-specific features.
- Modality-aware augmentation improves robustness by respecting the physical meaning of each channel.
- The best performance comes from combining architecture design and data-centric augmentation.

# What I Learned from This Independent Study

- Multimodal learning requires more than simply stacking input channels.
- Data preprocessing and augmentation must respect the physical meaning of each modality.
- Controlled ablation studies are necessary to verify where improvements come from.
- Model complexity and real inference speed do not always follow the same trend.
- The final work was developed into an ACM ICMR 2026 accepted paper.

# Thank you. Questions?

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